

Demonstration of Smart Parking System-

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PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 1
Enter vehicle type (car/bike/truck): 1. Car 2. Bike 3. Truck
1
Enter vehicle number: 101
Vehicle parked at slot: 1
Ticket ID: 1706234378

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 1
Enter vehicle type (car/bike/truck): 1. Car 2. Bike 3. Truck
3
Enter vehicle number: 1177
Vehicle parked at slot: 2
Ticket ID: 2046562095

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 2
Enter Ticket ID: 1706234378
Vehicle exited. Fee: 20.0

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 4
Exiting system...
PS C:\Users\Shreshta Ganapuram\OneDrive\Attachments\Desktop\2. OOP Project>
```

The application is a **menu-driven console program** that continuously interacts with the user until termination.

At each step, the user is prompted to choose one of the following options:

- **1. Park Vehicle**
- **2. Exit Vehicle**

- **3. Show Available Slots**
- **4. Exit Application**

When the user selects **option 4**, the application terminates.

Parking a Vehicle

- The user enters the **vehicle type** (Car/Bike/Truck) and **vehicle number**.
- The system allocates the **nearest available parking slot** using the implemented strategy.
- A **unique ticket ID** is generated and displayed.

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1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 1
Enter vehicle type (car/bike/truck): 1. Car  2. Bike  3. Truck
1
Enter vehicle number: 107
Error: Parking is full!

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 3
Available Slots:

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 1
Enter vehicle type (car/bike/truck): 1. Car  2. Bike  3. Truck
3
Enter vehicle number: 1199
Error: Parking is full!

```

Exiting a Vehicle

- The user enters the **ticket ID**.
- The system validates the ticket and:
 - Frees the allocated parking slot
 - Calculates the **parking fee** based on vehicle type and duration
- If an **invalid ticket ID** is entered, an appropriate error message is shown.

Viewing Available Slots

- Displays all currently **unoccupied parking slot numbers**.
- If all slots are occupied, the system indicates that **no slots are available**.

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===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 1
Enter vehicle type (car/bike/truck): 1. Car 2. Bike 3. Truck
55
Invalid type!
```

```
===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 1
Enter vehicle type (car/bike/truck): 1. Car 2. Bike 3. Truck
1
Enter vehicle number: 109
Vehicle parked at slot: 1
Ticket ID: 1706234378

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 2
Enter Ticket ID: 17060000
Invalid Ticket!
```

Exception Handling Cases

The system includes robust exception handling to manage real-world scenarios:

- **✗ Parking Full**
 - When all parking slots (e.g., 5 slots) are occupied, attempting to park another vehicle throws a custom exception.
 - The system displays a message indicating that parking is full.
- **✗ Invalid Vehicle Type**
 - If the user enters an unsupported vehicle type, an error is raised and handled gracefully.
- **✗ Invalid Ticket ID**
 - If the entered ticket ID does not exist, the system prevents the operation and shows an error message.
- **✗ Incorrect Input Format**
 - Handles unexpected or invalid user inputs without crashing the program.

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===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 3
Available Slots:
3 4 5

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 1
Enter vehicle type (car/bike/truck): 1. Car  2. Bike  3. Truck
2
Enter vehicle number: 222
Vehicle parked at slot: 3
Ticket ID: 1587487668

===== SMART PARKING SYSTEM =====
1. Park Vehicle
2. Exit Vehicle
3. Show Available Slots
4. Exit Program
Enter choice: 4
Exiting system...
PS C:\Users\Shreshta Ganapuram\OneDrive\Attachments\Desktop\2. OOP Project>

```

Continuous Operation

- The system runs in a loop, allowing multiple parking and exit operations.
- Once a vehicle exits, its slot becomes available again for new vehicles.